

Ecommerce Sales Dashboard

Project objective: Ecommerce Sales Analysis & Dashboard (Excel)

Objective: Clean, integrate, and analysis two years of ecommerce transaction data (Oct 2023 to 2024) to find out which products, customers, stores, and regions drive revenue. Present the findings in an interactive Excel dashboard that supports sales and marketing decisions, and forecast future trends.

Dataset Description: The E-Commerce Sales Dataset contains transactional sales data covering customers, products, sales quantities, unit prices, discounts, and total sales amounts. The dataset is organized into different dimensions, such as Sales, Product, and Customer details. It can be used to analysis sales performance, product trends, customer behaviour, discounts, and revenue patterns.

Column Description:

Column Name	Description
Month	Provides order data
Customer id	Provides the customer id
Sales id	Provides the sale id for the give below
Unit price	Provides the unit price for the each product
Region	Provides the region for each state
City	Provides the city name
Gender	Provides the gender male or female
Loyalty level	Provides the loyalty level for each customer
Category	Provides the category for the product the base
Discount	Provides the discount gives to the customer

Data cleaning used formal

Converting the Raw Data into Table

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donalnton	North Carolina	USA	Unknown
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Unknown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	Unknown
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum
CUST1029	David Arnold	54	Female	East Jennifer	California	USA	Silver
CUST1030	Jessica Acevedo	64	Female	Amybury	Pennsylvania	USA	Bronze
CUST1031	Sarah Vance	41	Male	Johnsonville	Oregon	USA	Silver
CUST1032	Mike Jennings	48	Female	Fostermouth	Arizona	USA	Gold
CUST1033	Robert Jones Dds	25	Male	Tristanton	Oklahoma	USA	Platinum
CUST1034	Christopher Olson	27	Female	West Richard	Maryland	USA	Platinum
CUST1035	Rachel Myers	52	Male	Robertberg	North Dakota	USA	Silver
CUST1036	Thomas Phillips	20	Female	Luisfurt	Georgia	USA	Gold
CUST1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA	Unknown
CUST1038	Deanna Simpson	35	Female	Brownport	North Dakota	USA	Bronze
CUST1039	Tonya Reynolds	55	Female	Darleneton	Michigan	USA	Gold
CUST1040	Eric Vega	50	Male	Williamsonland	Kansas	USA	Gold
CUST1041	James Gibbs	21	Male	Mooneyshire	Alaska	USA	Platinum
CUST1042	William Martin	19	Male	Ryanburgh	Connecticut	USA	Platinum

customer id typos using the find and replace

```
proper(trim(clean(a2)))
```

column b,e,f

```
proper(trim(clean(a2)))
```

loyalty level using blanks values replace in unknown

```
if(or(isblanks(h2),h2=0),"unknown",0)
```

	SALE_ID	Order Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit Price	Discount	Payment Type	Total Amount
2	SALE0	07/07/24	CUST1409	PROD97	STORE3	1.00	\$218.04	0.09	Credit Card	198.4164
3	SALE1	25/12/23	CUST1467	PROD74	STORE17	3.00	\$104.47	0.27	PayPal	228.7893
4	SALE2	03/06/24	CUST1056	PROD49	STORE8	4.00	\$914.05	0.25	Credit Card	2742.15
5	SALE3	29/11/23	CUST1346	PROD76	STORE8	1.00	\$921.51	0.16	COD	774.0684
5	SALE4	06/01/24	CUST1493	PROD59	STORE5	3.00	\$745.30	0.18	Credit Card	1833.438
7	SALE5	26/07/24	CUST1116	PROD15	STORE7	3.00	\$147.01	0.19	COD	357.2343
8	SALE6	12/07/25	CUST1016	PROD56	STORE4	2.00	\$785.31	0.15	PayPal	1335.027
9	SALE7	18/11/23	CUST1152	PROD70	STORE8	4.00	\$342.39	0.23	PayPal	1054.5612
0	SALE8	04/05/25	CUST1187	PROD99	STORE5	1.00	\$448.25	0.07	PayPal	416.8725
1	SALE9	31/07/25	CUST1464	PROD33	STORE0	2.00	\$873.17	0.23	PayPal	1344.6818
2	SALE10	13/02/24	CUST1065	PROD24	STORE14	1.00	\$278.63	0.07	Credit Card	259.1259
3	SALE11	24/05/25	CUST1320	PROD39	STORE1	4.00	\$286.92	0.17	Credit Card	952.5744
4	SALE12	24/08/24	CUST1237	PROD88	STORE4	4.00	\$644.58	0.01	Credit Card	2552.5368
5	SALE13	06/08/25	CUST1443	PROD34	STORE16	2.00	\$316.90	0.3	PayPal	443.66
6	SALE14	21/06/25	CUST1348	PROD24	STORE15	4.00	\$403.47	0.29	PayPal	1145.8548
7	SALE15	10/12/24	CUST1435	PROD36	STORE9	1.00	\$871.34	0.04	COD	836.4864
8	SALE16	13/09/25	CUST1276	PROD65	STORE19	2.00	\$451.98	0.25	Credit Card	677.97
9	SALE17	11/12/24	CUST1079	PROD27	STORE4	2.00	\$502.69	0.23	Credit Card	774.1426
0	SALE18	13/08/24	CUST1094	PROD39	STORE4	2.00	\$559.23	0.25	COD	838.845
1	SALE19	18/03/24	CUST1239	PROD60	STORE13	4.00	\$398.89	0.04	COD	1531.72224
2	SALE20	05/07/25	CUST1144	PROD72	STORE17	3.00	\$986.04	0.27	COD	2159.4276
3	SALE21	04/06/24	CUST1396	PROD43	STORE18	1.00	\$589.56	0.22	Credit Card	459.8568
4	SALE22	12/04/24	CUST1476	PROD69	STORE10	3.00	\$979.42	0.03	PayPal	2850.1122
5	SALE23	23/06/24	CUST1368	PROD12	STORE8	1.00	\$719.28	0.08	Credit Card	661.7376
6	SALE24	08/09/25	CUST1186	PROD67	STORE6	3.00	\$77.60	0.07	Credit Card	216.504
7	SALE25	17/09/24	CUST1381	PROD5	STORE3	4.00	\$852.71	0.14	PayPal	2933.3224
8	SALE26	03/01/24	CUST1213	PROD98	STORE8	4.00	\$156.16	0	Credit Card	624.64
9	SALE27	22/02/25	CUST1200	PROD77	STORE8	3.00	\$815.49	0.17	Credit Card	2030.5701
0	SALE28	23/09/25	CUST1148	PROD13	STORE2	2.00	\$74.21	0.11	PayPal	132.0938
1	SALE29	11/01/24	CUST1348	PROD99	STORE3	1.00	\$451.67	0.15	Credit Card	383.9195
2	SALE30	18/10/23	CUST1028	PROD5	STORE19	1.00	\$429.38	0.13	PayPal	373.5606
3	SALE31	28/10/24	CUST1412	PROD53	STORE12	2.00	\$148.96	0.03	COD	288.9824
4	SALE32	01/11/24	CUST1398	PROD75	STORE8	4.00	\$313.83	0.08	Credit Card	1154.8944
5	SALE33	24/11/24	CUST1453	PROD12	STORE6	2.00	\$717.31	0.15	Credit Card	1219.427
6	SALE34	16/10/23	CUST1344	PROD79	STORE1	1.00	\$873.22	0.18	Credit Card	716.0404
7	SALE35	09/08/25	CUST1218	PROD16	STORE3	4.00	\$109.36	0.19	COD	354.3264
8	SALE36	09/12/24	CUST1086	PROD5	STORE7	3.00	\$559.52	0.16	PayPal	1409.9904
9	SALE37	07/02/24	CUST1492	PROD10	STORE17	3.00	\$121.76	0.27	PayPal	266.6544
0	SALE38	27/03/24	CUST1189	PROD33	STORE13	4.00	\$735.32	0.06	COD	2764.8032

unit price handling missing value using vlookup and average
VLOOKUP([@[Product_ID]],K:L,2,FALSE)
quantity blanks column using the find and repalce method
using 1
discoutn blanks column using the find and replace method
using 0
total amount is usign quantity and unit price *
[@Quantity]*[@[Unit_Price]]*(1-[@Discount])

<i>Total_Amount</i>		<i>Unit_Price</i>	
Mean	1085.156254	Mean	513.8903767
Standard Error	18.30980344	Standard Error	6.308112311
Median	866.6988	Median	522.43
Mode	#N/A	Mode	871.34
Standard Deviation	818.8393028	Standard Deviation	282.1073587
Sample Variance	670497.8038	Sample Variance	79584.56185
Kurtosis	0.067465901	Kurtosis	-1.190105562
Skewness	0.887468332	Skewness	-0.035794762
Range	3889.7838	Range	979.8
Minimum	18.459	Minimum	20.09
Maximum	3908.2428	Maximum	999.89
Sum	2170312.509	Sum	1027780.753
Count	2000	Count	2000
Largest(1)	3908.2428	Largest(1)	999.89
Smallest(1)	18.459	Smallest(1)	20.09
Confidence Level(95.0%)	35.90829701	Confidence Level(95.0%)	12.37116342

Used Descriptive Statistics using Toolpak

Product ID	Product Name	Category	Sub Category	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	\$112.42	416
PROD1	Men'S Graphic T-Shirt	Clothing	Tops	\$342.56	181
PROD2	Women'S Casual Hoodie	Clothing	Outerwear	\$254.04	352
PROD3	Men'S Running Jacket	Clothing	Outerwear	\$364.67	452
PROD4	Women'S Yoga Pants	Clothing	Bottoms	\$306.46	487
PROD5	Men'S Leather Belt	Clothing	Accessories	\$261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	\$312.80	350
PROD7	Led Table Lamp	Home	Lighting	\$218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	\$439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	\$495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	\$298.86	308
PROD11	Hd Smart Tv Stick	Electronics	Accessories	\$449.40	272
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	\$115.82	460
PROD13	Usb-C Fast Charger	Electronics	Accessories	\$242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	\$48.01	439
PROD15	4K Home Projector	Electronics	Appliances	\$465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	\$337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	\$144.53	367
PROD18	Women'S Denim Jeans	Clothing	Bottoms	\$254.04	343
PROD19	Knee Support Brace	Sports	Equipment	\$21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	\$131.51	272
PROD21	Multi-Purpose Storage Box	Home	Storage	\$123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	\$484.62	432
PROD23	Men'S Formal Shirt	Clothing	Tops	\$254.04	360
PROD24	Foldable Camping Chair	Sports	Outdoor	\$166.06	340
PROD25	Women'S Scarf Set	Clothing	Accessories	\$174.02	268
PROD26	Wireless Bluetooth Speaker	Electronics	Audio	\$322.41	213
PROD27	Decorative Floor Vase	Home	Decor	\$254.04	378
PROD28	Sports Stopwatch Timer	Sports	Accessories	\$38.76	137
PROD29	Fabric Sofa Throw Blanket	Home	Decor	\$218.48	276
PROD30	Stainless Steel Cutlery Set	Home	Kitchen	\$19.52	57
PROD31	Basketball	Sports	Team Sports	\$204.39	281
PROD32	Vitamin C Face Serum	Beauty	Skincare	\$282.82	244

Using the formula :

=if(or(isblank(e2),e=0),average(e:e),e2)

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Gender	loyaltylevel	category	sub_category	region	store_type	product_name
SALE0	07/07/24	CUST1409	PROD97	STORE3	1.00	\$218.04	0.09	Credit Card	198.4164	Male	Gold	Sports	Team Sports	North	Outlet	Cricket Bat
SALE1	25/12/23	CUST1467	PROD74	STORE17	3.00	\$104.47	0.27	PayPal	228.7893	Female	Gold	Electronics	Audio	North	Flagship	Bluetooth Car Audio System
SALE2	03/06/24	CUST1056	PROD49	STORE8	4.00	\$914.05	0.25	Credit Card	2742.15	Female	Gold	Beauty	Fragrance	West	Online	Men'S Roll-On Perfume
SALE3	29/11/23	CUST1346	PROD76	STORE8	1.00	\$921.51	0.16	COD	774.0684	Female	Platinum	Electronics	Computer	West	Online	Wireless Mouse
SALE4	06/01/24	CUST1493	PROD59	STORE5	3.00	\$745.30	0.18	Credit Card	1833.438	Male	Silver	Beauty	Haircare	North	Outlet	Men'S Hair Styling Gel
SALE5	26/07/24	CUST1116	PROD15	STORE7	3.00	\$147.01	0.19	COD	357.2343	Female	Platinum	Electronics	Appliances	South	Online	4K Home Projector
SALE6	12/07/25	CUST1016	PROD56	STORE4	2.00	\$785.31	0.15	PayPal	1335.027	Female	Bronze	Home	Decor	South	Online	Premium Blanket Throw
SALE7	18/11/23	CUST1152	PROD70	STORE8	4.00	\$342.39	0.23	PayPal	1054.5612	Female	Platinum	Home	Storage	West	Online	Kitchen Storage Rack
SALE8	04/05/25	CUST1187	PROD99	STORE5	1.00	\$448.25	0.07	PayPal	416.8725	Female	Platinum	Home	Kitchen	North	Outlet	Stainless Steel Cookware Set
SALE9	31/07/25	CUST1464	PROD33	STORE0	2.00	\$873.17	0.23	PayPal	1344.6818	Female	Platinum	Home	Decor	East	Flagship	Modern Wall Clock
SALE10	13/02/24	CUST1065	PROD24	STORE14	1.00	\$278.63	0.07	Credit Card	259.1259	Female	Silver	Sports	Outdoor	South	Outlet	Foldable Camping Chair
SALE11	24/05/25	CUST1320	PROD39	STORE1	4.00	\$286.92	0.17	Credit Card	952.2744	Female	Bronze	Electronics	Mobile	North	Flagship	Portable Power Bank
SALE12	24/08/24	CUST1237	PROD88	STORE4	4.00	\$644.58	0.01	Credit Card	2552.5368	Female	Gold	Sports	Accessories	South	Online	Sports Wrist Sweatbands
SALE13	06/08/25	CUST1443	PROD34	STORE16	2.00	\$316.90	0.3	PayPal	443.66	Male	Silver	Electronics	Computer	West	Flagship	Gaming Mechanical Keyboard
SALE14	21/06/25	CUST1348	PROD24	STORE15	4.00	\$403.47	0.29	PayPal	1145.8548	Male	Silver	Sports	Outdoor	North	Flagship	Foldable Camping Chair
SALE15	10/12/24	CUST1435	PROD36	STORE9	1.00	\$871.34	0.04	COD	836.4864	Female	Silver	Clothing	Tops	South	Flagship	Women'S Summer Dress
SALE16	13/09/25	CUST1276	PROD65	STORE19	2.00	\$451.98	0.25	Credit Card	677.97	Male	Gold	Sports	Fitness	East	Online	Adjustable Dumbbells 20Kg
SALE17	11/12/24	CUST1079	PROD27	STORE4	2.00	\$502.69	0.23	Credit Card	774.1426	Female	Platinum	Home	Decor	South	Online	Decorative Floor Vase
SALE18	13/08/24	CUST1094	PROD39	STORE4	2.00	\$559.23	0.25	COD	838.845	Female	Gold	Electronics	Mobile	South	Online	Portable Power Bank
SALE19	18/03/24	CUST1239	PROD60	STORE13	4.00	\$398.89	0.04	COD	1531.72224	Female	Platinum	Beauty	Fragrance	East	Flagship	Luxury Perfume Spray
SALE20	05/07/25	CUST1144	PROD72	STORE17	3.00	\$986.04	0.27	COD	2159.4276	Female	Bronze	Sports	Fitness	North	Flagship	Professional Jump Rope
SALE21	04/06/24	CUST1396	PROD43	STORE18	1.00	\$589.56	0.22	Credit Card	459.8568	Female	Silver	Electronics	Computer	East	Flagship	Usb External Hard Drive
SALE22	12/04/24	CUST1476	PROD69	STORE10	3.00	\$979.42	0.03	PayPal	2850.1122	Female	Gold	Clothing	Bottoms	East	Online	Men'S Sports Shorts
SALE23	23/06/24	CUST1368	PROD12	STORE8	1.00	\$719.28	0.08	Credit Card	661.7376	Female	Platinum	Beauty	Haircare	West	Online	Hair Strengthening Shampoo
SALE24	08/09/25	CUST1186	PROD67	STORE5	3.00	\$77.60	0.07	Credit Card	216.504	Male	Silver	Electronics	Accessories	East	Online	Smart Led Light Strip
SALE25	17/09/24	CUST1381	PROD5	STORE3	4.00	\$852.71	0.14	PayPal	2933.3224	Male	Silver	Clothing	Accessories	North	Outlet	Men'S Leather Belt
SALE26	03/03/24	CUST1213	PROD98	STORE8	4.00	\$156.16	0	Credit Card	624.64	Female	Platinum	Electronics	Audio	West	Online	Portable Bluetooth Speaker
SALE27	22/02/25	CUST1200	PROD77	STORE8	3.00	\$815.49	0.17	Credit Card	2030.5701	Male	Platinum	Electronics	Appliances	West	Online	WiFi Mesh Router
SALE28	23/09/25	CUST1148	PROD13	STORE2	2.00	\$74.21	0.11	PayPal	132.0938	Female	Bronze	Electronics	Accessories	South	Outlet	Usb-C Fast Charger
SALE29	11/01/24	CUST1348	PROD99	STORE3	1.00	\$451.67	0.15	Credit Card	383.9195	Male	Silver	Home	Kitchen	North	Outlet	Stainless Steel Cookware Set
SALE30	18/10/23	CUST1028	PROD5	STORE19	1.00	\$429.38	0.13	PayPal	373.5606	Female	Platinum	Clothing	Accessories	East	Online	Men'S Leather Belt
SALE31	28/10/24	CUST1412	PROD53	STORE12	2.00	\$148.96	0.03	COD	288.9824	Female	Gold	Clothing	Bottoms	North	Online	Men'S Track Pants
SALE32	01/11/24	CUST1398	PROD75	STORE8	4.00	\$313.83	0.08	Credit Card	1154.8944	Male	Bronze	Clothing	Footwear	West	Online	Women'S Ankle Boots

using the vlookup formal combine sheet in gender column for analysis
for pivot tabless

VLOOKUP([@[Customer_ID]],Customer_Dim!A:D,4,FALSE)

using the vlookup formal combine sheet in loyalty level column for analysis
for pivot tabless

VLOOKUP([@[Customer_ID]],Customer_Dim!A:H,8,FALSE)

using the vlookup formal combine sheet in poduct id category column for
analysis
for pivot tabless and also sub category

VLOOKUP([@[Product_ID]],Product_Dim!A:D,3,FALSE)

VLOOKUP([@[Product_ID]],Product_Dim!A:D,4,FALSE)

using the vlookup formal combine sheet in region column for analysis
which region highest sale for pivot tables

VLOOKUP([@[Store_ID]],Store_Dim!A:D,3,FALSE)

using the vlookup formal combine sheet in region column for analysis
for which store type sale

VLOOKUP([@[Store_ID]],Store_Dim!A:E,5,FALSE)

Visualization and Insights Arrived with screenshot of visualization

Report/Dashboard screenshot:

1. Descriptive Analysis — *What happened?*

What is the average sales value per order?

Which product category has the highest sales?

Which region/store generated the highest sale?/

2. Diagnostic Analysis — *Why did it happen?*

Why did one product category generate higher sales than another?

Which products contributed most to the difference in sales?

3. Predictive Analysis — *What is likely to happen?*

Based on historical sales, can we forecast future revenue?

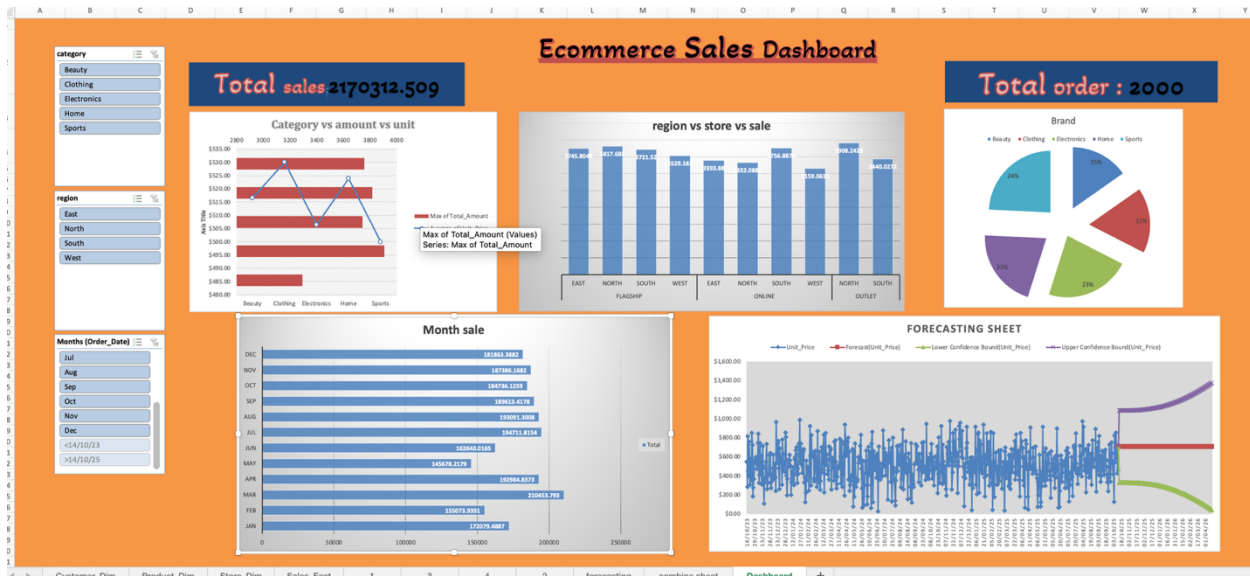
Which stores/regions are likely to continue performing well?

4. Prescriptive Analysis — *What should we do?*

Which products should the company promote to increase sales?

What discount strategy could improve sales?

Dashboard Image:



Final 4 type of insight for storyline :

We have 2000 sale order placed order by customer and total sale amount \$ 2170312.509 in sale amount and bought 4,960 total items , the average order was about \$1,087. Out of all category sport and electronics and home products. The payment method is customer have credit card and paypal and also (COD) cash o delivery . Electronics and Sports look like they will keep growing and remain customer favorites. Put more advertising and stock into our top-selling categories like Sports and Home . West region only has 12% of sales, so we should run special promos there to catch up with the North and East.

Conclusion:

This project taught me that data isn't just a scary spreadsheet, it is actually a story about people shopping, buying, and helping a

business grow. I am so proud of building my first dashboard and finding these insights.

Thank you